Selma Marrakchi

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Grounded in neuroscience and biomedical engineering, with hands-on research experience and years of direct work with clients and patients. I am driven by lifelong learning and a strong dedication to helping others grow and thrive, aiming to integrate practice and research in meaningful ways that advance the field.

Education

Columbia University — Biomedical Engineering, MS May 2023
Bioinformatics & Machine Learning Specialization, Advanced Research Specialization GPA: 3.98
Courses: Statistical ML, Data Science, Bioinformatics, Biomedical Design, Computational Modeling, Genomics

University of California, Los Angeles — Neuroscience, BS Sept 2017
With an emphasis on neurophysics, neural computing in learning and memory, and psychology GPA: 3.42
Courses: Multivariate Calculus, Statistics, Thermo/Fluid Dynamics, Molecular Biology/Genetics, Chemistry, Biochemistry

Experience

Uncountable Inc. — ***Scientific Implementation Associate*** Jan 2025 - Present
- Leveraging scientific background to implement powerful solutions for industry-leading customers across a wide variety of sectors.
- Translating experimental workflows into scalable, structured digital systems to speed up RD, optimize processes, and drive business development impact.

New York Genome Center — ***Associate Research Scientist II*** Sept 2023 - Present
- Used spatial transcriptomic and computational tools to investigate the microenvironment characterizing colorectal cancer tumors.
- Built and optimized RNA sensors to deplete noradrenergic receptor subtypes in neuronal populations in-vivo to study complex neural computation of behavior in mice.

The Chemical Genomics Lab, Columbia University — ***Graduate Researcher*** Jan 2022 - June 2023
- Optimized lentivirus production and lentivirus-mediated transduction protocols on HEK293 cell lines.
- Induced KO mutations in rat neural cells targeting 5 Alzheimer's disease-related genes through lentiCRISPR technique.

Behavior Guidance Group — ***BACB-Registered Behavior Technician*** Dec 2019 - Nov 2020
- Implemented behavior analytic therapy interventions in school and at-home settings, accruing over 1500 hours of 1:1 experience.
- Optimized treatment plans for 5 clients with case supervisors through extensive daily data collection and analysis.

Vital Head & Spinal Care — ***Lead EEG and Neurofeedback Technician*** May 2019 - Nov 2019
- Spearheaded transition into the office's 2 new neurofeedback stations and EEG recording room; set up all hardware and software.
- Administered and managed treatment plans for 75+ patients under the supervision of Dr. Hill (PBI) and Dr. Licata (VHSC).
- Trained 2 new hires in EEG recording and NFB therapy to full proficiency in NeuroGuide, EEGer and BioExplorer softwares.

Peak Brain Institute — ***Jr. EEG and Neurofeedback Technician*** Jan 2018 - May 2019
- Recorded 350+ clinical-quality wet EEGs and processed data for artifacts to generate qEEG brain maps.
- Administered NFB/HEG therapy to 16+ patients daily; set up and optimized treatment interfaces based on real-time EEG data.
- Collaborated on 2 research projects. Trained 5 interns. Maintained patient charts, inventory, and all hardware and equipment.

Leadership and Achievements

2022-2023 Advanced Research Fellowship, Columbia SEAS
Received a scholarship worth $30k awarded to 2-3 students per academic year based on demonstrated research excellence.

2022 Biomedical Design & Innovation Awards - Best Business Strategy
Developed a portable stroke detection prototype, coupling hardware (wearable device) and software (AI model + telehealth platform) components, to expediate treatment and significantly improve accessibility and affordability of care.

Graduate Organization for Biomedical Engineers, Columbia SEAS — Internal VP Jan 2022 - Jan 2023

Engineering Graduate Student Council, Columbia SEAS — Lead Dept Representative Jan 2022 - Jan 2023

Skills and Interests

Languages: French (Native), Arabic (Fluent), Spanish (Professional Working Proficiency).
Interests: Guitar, drums, violin, music composition, climbing, boxing, chess.